

This article was downloaded by: [University of Chicago Library]

On: 19 August 2013, At: 01:37

Publisher: Routledge

Informa Ltd Registered in England and Wales Registered Number: 1072954 Registered office: Mortimer House, 37-41 Mortimer Street, London W1T 3JH, UK



Review of Social Economy

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/rrse20>

The Neoclassical and Post-Keynesian Research Programs: The Methodological Issues

Elba K. Brown ^a

^a University of Texas at El Paso

Published online: 02 Nov 2006.

To cite this article: Elba K. Brown (1981) The Neoclassical and Post-Keynesian Research Programs: The Methodological Issues, Review of Social Economy, 39:2, 111-132, DOI: [10.1080/758538453](https://doi.org/10.1080/758538453)

To link to this article: <http://dx.doi.org/10.1080/758538453>

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden. Terms & Conditions of access and use can be found at <http://www.tandfonline.com/page/terms-and-conditions>

REVIEW OF SOCIAL ECONOMY

VOLUME XXXIX

OCTOBER, 1981

NUMBER 2

THE NEOCLASSICAL AND POST-KEYNESIAN RESEARCH PROGRAMS: THE METHODOLOGICAL ISSUES*

By ELBA K. BROWN**
University of Texas at El Paso

The debate about the Keynesian revolution rages on and on, seemingly never to end. It continues in various forms, sometimes referred to as the Cambridge controversy, sometimes as the post-Keynesian/neoclassical debate. By whatever name it is called, the essence of the debate is whether or not the post-Keynesian approach is a viable alternative to the well-established neoclassical view as an explanation of an aggregated economic process. Are these in fact alternative and contradictory research programs? If so, how should economists rationally choose one as the better explanation? The argument presented here is that two contradictory research programs do exist, resulting in quite different theoretical frameworks or approaches used to "explain" a dynamic macroeconomic process. The resolution concerning which of these alternative approaches is the better "explanation" involves the resolution of important methodological issues within the discipline of economics. Until there is consensus regarding what constitutes a "better" theory of a "better" research program, there can be no consensus in the science of economics concerning the appropriate approach.

A few words of caution are in order with respect to the interpretation of what follows. First, the concept of a research program is a framework for analyzing the basic contradictions between the two approaches. That concept and the difficulties in using it are discussed below. Second, in focusing in on two different approaches for ex-

*0034-6764/81/1001-111/\$1.50/0.

**Assistant Professor of Economics.

plaining a macroeconomic system, it is impossible to easily categorize each and every economist as being in either one camp or another. Specifically, there are many areas where such classification cannot be made. For example, individuals working in general equilibrium theory are often more concerned with showing why a free market economy is *not* likely to be coordinated at a unique optimum rather than an analysis of how a coordinated outcome is achieved.

There is by now a long and fairly imposing line of economists from Adam Smith to the present who have sought to show that a decentralized economy motivated by self-interest and guided by price signals would be compatible with a coherent disposition of economic resources that could be regarded, in a well-defined sense, as superior to a large class of possible alternative dispositions. . . . The proposition having been put forward and very seriously entertained, it is important to know not only whether it *is* true, but also whether it *could* be true. A good deal of what follows is concerned with this last question . . . [Arrow and Hahn, p. vii]

One is tempted to classify Kenneth Arrow and F. H. Hahn as neo-classical because of their use of the neoclassical general equilibrium framework. Yet their purpose is more closely associated with what will be described below as the "hard core" of the post-Keynesian approach. Third, one should not confuse the post-Keynesian/neoclassical debate with the monetarist/Keynesian (American) debate. And, finally, the purpose of this paper is not to debate all of the theoretical issues involved in this controversy, but rather to show that such debates are meaningless in the absence of any consensus with respect to the methodological and epistemological issues involved.

THE RESEARCH PROGRAM

The concept of a research program was developed by Imre Lakatos in an attempt to explain the evolution of a scientific discipline. [See Lakatos.] A research program is defined as a series of theories which share a common "hard core" belief. All of the theories in the program must be consistent with the "hard core" belief and the "hard core" of the program is "irrefutable" by the methodological decision of its protagonists. [Lakatos, p. 135] The postulates of the "hard core" are accepted, and the "negative heuristic" forbids testing or questioning of the "hard core" principles. It is important to understand that

these "hard core" principles need not be well articulated. Indeed, the act of articulating the "hard core" principles may expose them to criticism and question.¹ In particular, one should not confuse the axioms of a theoretical framework with the "hard core" beliefs with which those axioms must be consistent. If the "hard core" beliefs are stated as the axioms of a theory or group of theories, the "hard core" is exposed to criticism and testing. As a research program progresses, the axioms of the theories in that program may have to change in order to protect the "hard core" belief. Rules of procedure in a scientific discipline are provided by the "positive heuristic"—a set of suggestions on how to change or develop the refutable part of the research program, the "protective belt."

The role of the research program in explaining the evolution of a science is that it defines the systematic framework within which each scientist works. The evolution of a discipline is described with respect to changes in the systematic framework. The rules for accepting such changes as scientific progress are discussed by Lakatos as rational decision-making within the discipline. The rules of demarcation set out by Karl Popper which provide grounds for selecting one theory over another are extended by Lakatos to apply to the selection of one research program over another. Lakatos' rules for appraising research programs are as follows:

A research program is either progressive or degenerating. It is *theoretically progressive* if each modification leads to new unexpected predictions and it is *empirically progressive* if at least some of these novel predictions are corroborated. It is always easy for a scientist to deal with a given anomaly by making suitable adjustments to his program. . . Such manoeuvres are *ad hoc*, and the program is degenerating unless they not only explain the given facts they were intended to explain but also predict some new fact as well. . . One research program supersedes another if it has excess truth content over its rival, in the sense that it predicts progressively all that its rival truly predicts and some more besides. [Lakatos and Zahar, p. 369]

Just how helpful these rules can be in choosing between two research programs, however, is a significant question. As Toulmin points out, what precisely is meant by "prediction" and "novel fact" is am-

¹ Because "hard core" principles are seldom made specific, application of the Lakatosian concept is particularly difficult. Often the "hard core" of a particular research program is specified quite differently from one writer to the next.

biguous. [Toulmin, pp. 384-91] Recognizing that objectively determined knowledge is an unattainable goal, Lakatos admits that a methodological decision is required to determine the "empirical base" with which a theory is tested. Before an agreement about whether a program is "progressive" or "degenerating" can be reached, there must also be an agreement about what constitutes a "prediction" of a "novel fact." While it seems reasonable to assume agreement among individuals within a given research program, there is at least a likelihood that no consensus will exist between rival research programs. Different views as to what constitutes "progressive content," both theoretically and empirically would involve these methodological decisions. Thus, it is possible to get two conflicting research programs within a scientific discipline, each describing and working within a different systematic framework with no objectively determined scientific ground upon which one might be selected as better than the other.

If a different systematic framework is associated with each research program, then what constitutes "explanation" in terms of one framework will not necessarily be accepted as "explanation" in the other. Toulmin claims that Lakatos' modification of his use of prediction in the reconstruction of the Copernican research program's victory over that of Ptolemy calls for replacing the word "prediction" with "explanation" if it is consistent with an accepted systematic view of the world. For example, the observed throwing motion made by a child is said to predict and explain the subsequent breaking of window-glass *because* such an explanation is consistent with the accepted view that such movement is associated with throwing an object, and that hard objects thrown against window-glass with certain properties will cause the glass to break. The prediction that the motion of the child would result in the breaking of the window is also explanation because it fits into a systematic framework of thought consistent with our view of the world. In Toulmin's interpretation of Lakatos, it is prediction in the sense of explanation within an accepted systematic framework which identifies the "progressivity" of a research program. If there is no agreement about the appropriate systematic framework, agreement on what constitutes better predictions or explanations cannot easily be reached.

The "hard core" beliefs of the research program must be consistent with the view of the world which defines the systematic framework accepted by the protagonists of that program. Two contradictory re-

search programs would have different frameworks, each consistent with its own "hard core." Thus, competing research programs would involve different notions about what constitutes predictive and explanatory power of a given theory. If we can identify the different "hard core" beliefs and the resulting differences in the systematic framework, we can better understand the nature of the debate between neoclassicists and the post-Keynesians.

THE RESEARCH PROGRAM EVIDENT IN THE WORKS OF KEYNES

There are different possible specifications of the neoclassical "hard core." The specification made by Spiro Latsis is most acceptable from my point of view. It is this specification I shall use as a comparison to the "hard core" evident in the works of Keynes and in the works of the post-Keynesians. Latsis describes the "hard core" of the neoclassical research program as follows:

- (i) Decision-makers have correct knowledge of the relevant features of their economic situation.
- (ii) Decision-makers *prefer* the best available alternative given their knowledge of the situation and of the means at their disposal.
- (iii) Given (i) and (ii), situations generate their internal "logic" and decision-makers *act appropriately to the logic of their situation*.
- (iv) Economic units and structures display stable, coordinated behavior. [Latsis, p. 22]

This notion of "stable and coordinated behavior" becomes in neoclassical macrotheory an ideal toward which the economic process is said to move, with policy prescriptions derived as those policies which would facilitate that movement. The coordinated outcome is implicitly accepted in neoclassical macrotheory and explicitly accepted in neoclassical welfare theory as a position toward which the economic process *should* move. The "efficient" allocation of resources is defined in neoclassical economics as a Pareto optimal allocation—one from which there is no *potential* Pareto improvement. [Mishan, p. 274] Assuming the desirable distribution criteria or social welfare function is determined outside the "positive" neoclassical theory [Blaug, pp. 143-46], the neoclassical economists considers such an allocation an "ideal" even though ". . . it is not possible to say, in general, that more efficient production confers an unambiguous wel-

fare gain in the form of a potential Pareto improvement on the community or that a production optimum is better in the sense for the community than a nonoptimum." [Mishan, p. 348]

The Keynesian "hard core" contradicts the neoclassical "hard core" in two respects. First, decision-makers cannot have correct knowledge of the relevant features of their economic situation and, second, as a result, the system is not inherently stable and coordinated. The actual economic process does not automatically move toward an "efficient" allocation or production optimum. These differences are reflected in the framework of analysis used by each group.

The research program evident in the works of Keynes had the following characteristics. First, he did not hold that decision-makers have correct knowledge of the relevant features of their economic situation. In the *Treatise on Probability* Keynes clearly presents his epistemological views. "Knowledge" of the future can only be gained via induction from past experiences. What we can know is "direct" knowledge. Knowledge by induction is "indirect knowledge." We may have a "rational degree of belief" in the "indirect knowledge" of a proposition by perceiving a logical relation between the proposition and "direct knowledge." This logical relation Keynes termed probability.

Decisions between alternative courses of action must involve knowledge of the future. We must have, or believe ourselves to have, knowledge of the future consequences of each alternative in order to choose one alternative course of action over another. It appears that Keynes would argue that such "knowledge" of the future must be held with only a probable degree of certainty. But, Keynes was careful to stipulate the conditions under which a probability could be calculated. These conditions are seldom met in the course of everyday activity. Thus, in general, such probability calculations cannot be made.

By uncertain knowledge, let me explain, I do not mean merely to distinguish what is known for certain from what is only probable. . . . About these matters there is no scientific basis to form any calculable probability whatever. We simply do not know. [Keynes, "The General Theory of Employment," in 1973b, p. 114]

Not only can probability calculations not be made in most situations, but comparisons of probability calculations cannot be made. Thus, for decisions made with reference to the future consequences of current actions, correct knowledge cannot be held.

Second, as a result of this lack of correct knowledge, Keynes would not accept the proposition that economic units and structures display stable, coordinated behavior. Coordination, in the sense that pre-reconciled plans yield a unique single-exit decision, does not exist. Single-exit decision situations are defined as "situations where one best course of action is uniquely prescribed by the structure of the situation according to some liberal conception of rational behavior." [Latsis, p. 19] Conversely, a multiple-exit decision situation "is physically and situationally open." It is physically open because the subsequent action does not appear to be narrowly circumscribed by available physical or physiological laws and conditions, and it is situationally open in the sense that the socioeconomic environment "does not (given the decision-maker's aims, interest, and objective) uniquely determine any particular course of action." [Latsis, p. 22]

The Keynesian "hard core," then, can be presented as follows:

- (i) Decision-makers prefer the best available alternatives given their knowledge of the situation and the means at their disposal.
- (ii) Decision-makers behave in a "rational" way, i.e., they use logic in their reasoning.
- (iii) Decision-makers do not have correct knowledge of situation with respect to alternative courses of action. Although decision-makers are "rational," there is no information or "knowledge" about some alternatives with which a "rational" decision can be made.
- (iv) Given (iii), economic units and structures do not display coordinated behavior. Expectations are not realized, and markets do not necessarily clear. A unique or ideal equilibrium does not result from individual decision-making. Multiple-exit decision situations are characteristic of the economic process.

"The positive heuristic" of a research program sets out the rules and procedures which the scientists follow. To determine what "positive heuristic" is associated with the Keynesian research program, we must investigate the methods used by Keynes and establish their consistency with the "hard core."

The most obvious feature, and perhaps the most controversial for our purposes, of Keynes' method was his use of comparative statics. Although, as Axel Leijonhufvud out, at the time "formal economic theory consisted almost altogether of static and comparative static

models with mathematical solutions only for 'coordinated states' of the system" [Leijonhufvud, p. 91], the relevant question is whether or not the method of comparative statics is consistent with the Keynesian "hard core." I would argue that claims of inconsistency can only be made if we impose on Keynes a neoclassical notion of equilibrium. For Keynes, as for Marshall, equilibrium was simply a balance or resting place from which the economy would not move unless a change or movement occurred in one of the change producing factors. These factors were assumed constant (long-term expectations) and fulfilled (short-term expectations) in order to describe in formal terms the "equilibrium." These types of "equilibria" are compared in *The General Theory*. But, if use of comparative statics is taken to require that the "equilibria" being compared are such that all markets must clear as a result of maximizing behavior or the part of perfectly informed individuals, then that method would be inconsistent with the Keynesian "hard core." Specifically, it would conflict with the notions that individuals cannot have complete knowledge of their economic situation, and that, as a result, coordination of the economic process in the form of such market-clearing is not generally the case.²

In order to describe an "equilibrium," it was necessary for Keynes to assume the psychological factors constant and to assume short-term expectations fulfilled. However, the resulting model with a determinate solution should not be interpreted as a single-exit decision situation in the sense that given the expectations, one and only one course of action would be prescribed by the "rational" behavior of individual decision-makers. But, in order for an "equilibrium" to obtain, it must be necessary that the actions taken by individuals do not change, at least in the aggregate. Notice the subtle but distinct difference. It is here that Keynes must move from the analysis of individual decision-making to the resulting actions taken as a result of some type of decision-making which may or may not be capable of analysis.

² One will note that I have described the market-clearing notion of equilibrium associated with what Leijonhufvud calls the Neo-Walrasian research program. Leijonhufvud's thesis is that the neoclassical and Neo-Walrasian research programs are not the same. I would argue, however, that the "hard core" set out by Latsis for the neoclassical program does apply to both groups. It could be argued that the equilibrium concept is a notion in the "protective belt" which had to be changed in order to be consistent with mathematical proofs that a unique single-exit decision situation could exist for the economy as a whole, thus protecting the neoclassical "hard core" from the Keynesian attack. This thesis, however, will not be pursued here.

Choices between alternative courses of action with respect to future consequences cannot be "rational," yet such decisions are made and actions result.⁸

The aggregation of such actions determines aggregate employment and income. If these are to remain the same, as they must in order for an "equilibrium" to obtain, there must be nothing happening to induce change in those actions. A change in long-term expectations would more than likely induce change in actions with respect to long-term alternatives. But, exactly what that change will be, given a change in long-term expectations, or even in what direction cannot be uniquely determined. The economic process is characterized by multiple-exit decision situations. Indeed, it is the analysis of such changes and of the different repercussions possible as a result that is important.

Time is embodied in Keynes' notion of uncertainty. Choices between alternative courses of action necessarily involve the future, as the decision can only be made with reference to the future outcome associated with each alternative. It is precisely these future outcomes about which we are uncertain, and for which no probability calculations or comparisons can be made. Factors which are partially determined by long-term expectations can never be completely determined by the existing conditions.

For example, the schedule of the marginal efficiency of capital depends partly on the existing quantity of equipment which is one of the given factors, but partly on the state of long-term expectation which cannot be inferred from the given factors. [Keynes, 1965, p. 246]

It is for this reason long-term expectations must be assumed constant in formulating an essentially "timeless" static model with a determinant solution. More importantly, however, it creates the multiple exit decision situation as a characteristic of the economic process. Thus, the Keynesian "equilibrium" model cannot be interpreted as a single-exit decision situation.

It must be made clear that Keynes was interested in analyzing changes in the economy, or the economic process over time. In the preface to *The General Theory*, Keynes criticizes his *Treatise on*

⁸It could be argued that such an approach is capable of greater empirical content than the orthodox approach, as the empirical observations are measurements or records of actions taken, not of an optimum desire, given certain assumptions.

Money because in it, "I failed to deal thoroughly with the effects of changes in the level of output." [Keynes, 1965, p. vii] But, he says of *The General Theory*, "This book, on the other hand, has evolved into what is primarily a study of the forces which determine changes in the scale of output and employment as a whole." [Keynes, 1965, p. vii]

The emphasis on change is closely tied to Keynes' notion of stability in the economic system. His view of the economic process was "that we oscillate, avoiding the gravest extremes of fluctuation in employment and in prices in both directions, round an intermediate position appreciably below full employment and appreciably above the minimum employment a decline below which would endanger life." [Keynes, 1965, p. 254] This apparent stability is due to certain special characteristics of the propensity to consume, the marginal efficiency of capital and the money rate of interest "about which we can safely generalize from experience, but which are not logically necessary." [Keynes, 1965, p. 249] Thus, for Keynes, the economy is constantly fluctuating between two extremes. However, the important differences between Keynes' and the neoclassical view is that these fluctuations are inherent in the system. It is not the case that the economic system is stable, and given a long enough period of time and no exogenous shock, full employment equilibrium will result. Change inducing characteristics are not exogenous; they are only exogenous to the static "equilibrium" model. Stability is discussed in terms of limits to those fluctuations, one limit of which is the case of full-employment. It is in this sense that Keynes contends that his theory is "a more general theory, which includes the classical theory with which we are familiar, as a special case." [Keynes, 1965, p. vii]

THE POST-KEYNESIAN RESEARCH PROGRAM

Our objective here is to analyze the differences in the neoclassical and post-Keynesian research programs.⁴ In what sense is current post-Keynesian theory an extension of the program evident in the works of Keynes?

⁴ By "post-Keynesian" we do not mean the American Keynesians. With the acceptance of the general equilibrium framework of IS-LM and the later extension of this with the addition of the labor market and the so-called Keynesian-neoclassical synthesis, American Keynesians have accepted the approach of the neoclassical framework. To show the insights of Keynes, the so-called "Keynesian constraints" must be added to the model. These simply have the effect of preventing adjustment to the full employment "market-clearing" equilibrium.

Consistent with the "hard core" belief that the economy is not automatically coordinated, the important questions for the post-Keynesians involve explanations of the actual observed behavior of the economic system. As Minsky points out, ". . . the explanation of systematic instability, rather than a tendency towards a stable equilibrium, became a key problem for economic analysis." [Minsky, p. 297] If one assumes that the economic process is not automatically coordinated at an optimal equilibrium, then explanations of movements of that process cannot be made in terms of movement toward a coordinated equilibrium, so the emphasis in post-Keynesian theory is placed on the discovery of what causes change.

. . . the impact of both past history and expectations about the future on current decisions that are bringing about the change must be taken into account. [Eichner and Kregel, p. 1294]

Post-Keynesians are thus ultimately concerned with the analysis of the economy in disequilibrium. [Eichner and Kregel, p. 1299]

An emphasis on change over time leads directly to an emphasis on growth and dynamics. However, growth cannot be explained by movement along an equilibrium path. Rather, the purpose is "to explain why, as the historical records bear strong witness, the expansion path of a free enterprise economy is likely to be so erratic." [Eichner and Kregel, p. 1296] What becomes important is to discover the main driving force of the economy and what endogenous features of the process lead to "systematic instability" or fluctuations.

The emphasis on growth and dynamics can be considered a logical extension of the Keynesian research program. Keynes was interested in what causes changes in investment and how those changes affect the level of output and employment. Recognizing that net investment means a change in the capital stock, it is logical that theories of growth should follow. Keynes used the method of comparative statics in order to isolate and discuss the major change-producing variables so that the fluctuations of the economic process could be explained. To achieve the same purpose, the post-Keynesians used the method of comparative dynamics. The difference in these methods is not substantial.

. . . but the main topic of *The General Theory* was the consequences of a *change* in the level of effective demand within a

short-period situation with given plant and available labor. The consequences of changing the stock of plant as investment matures hardly came into the story . . .

Once he (Keynes) had thrown off the incubus of Say's Law, the whole field of the long-period theory of accumulation remained to be explored. [Robinson, 1977, p. 1235]

The post-Keynesian emphasis on distribution can also be explained as consistent with the Keynesian "hard core." Acceptance of the marginal productivity theory of distribution would be inconsistent with the "hard core" if its acceptance requires the belief that an automatically coordinated equilibrium state exists. The post-Keynesian rejection of the marginal productivity theory of distribution rests on the criticism that the return to capital cannot be determined unless the capital-value aggregate could be specified. The prices of capital assets cannot be explained until it is assumed "that equilibrium not only now exists, but it has existed since the creation and will continue to exist until the final holocaust." [Minsky, 1977, p. 301] Thus, the marginal productivity theory of distribution involves the assumption that equilibrium in the sense of coordinated states must be at least possible. An alternative theory of distribution had to be developed within the Keynesian research program, i.e., consistent with the Keynesian "hard core."

This dissatisfaction with the traditional theory's explanation of the determination of factor prices, of the wage and profit rate, and thus of factor shares led to the search for a theory of distribution that was both logically sound and compatible with *The General Theory*. [Kregel, p. 423]

It should be noted that in addition to the impossibility of specifying a capital-value aggregate, the acceptance of Keynes' view that the level of employment could not be explained by marginal productivity of labor alone led to the abandonment of the marginal productivity theory of distribution. This, of course, is consistent with Keynes' notion of non-market clearing equilibrium.

One other characteristic which can be interpreted as consistent with the "hard core" of the Keynesian research program should be presented. The microeconomic base described by Eichner and Kregel as a distinguishing characteristic of post-Keynesian theory does *not* require the assumption of perfect information or foresight on the part of the individual decision-making units. The assumption of perfect competition is not required.

As noted above, Keynes' notion of equilibrium cannot be construed as necessitating market-clearing as in Walrasian models. In dynamic analysis, the notion of equilibrium is translated into a steady rate of expansion. Post-Keynesian growth theory begins by describing the conditions required for such a steady rate of expansion. However, it must be clear that such conditions do not include full-employment or clearing of all markets in the Walrasian sense. Post-Keynesian analysis is considered complete when "discretionary income must be equal to discretionary expenditures." [Eichner and Kregel, p. 1309]

This gives greater freedom to the analysis because, with only one set of flows that must be brought into balance before the chain of causal explanation can be considered complete, many other things—like the flow of those who would like to obtain employment or the desired portfolios of wealth holders—can remain out of balance. [Eichner and Kregel, p. 1301]

The significant difference between this view of equilibrium or steady expansion and the view requiring market-clearing is that the latter view associates simultaneity of determination in the model with causality.

One of the uses of static equilibrium has been to describe the system with a set of simultaneous equations, yielding a determinate system. The Keynesian system is not completely determined by the "givens" because the "independent" variables are those which can never be inferred from the given variables. For the post-Keynesians, as for Keynes, the notion of causality "is not the same as simultaneity of determination." [Eichner and Kregel, p. 1295]

... a most reliable sign of distortion shows up any time the original Keynes' "clear-cut results" are obfuscated by the imposition of interdependences which transform Keynes' causally ordered relations into a system of simultaneous equations. [Pasinetti, p. 46]

Post-Keynesians do not call for abandonment of the equilibrium concept. However, caution in interpreting it as a point toward which the economy should or does move is expressed.

The concept of equilibrium, of course, is an indispensable tool of analysis. ... But to use the equilibrium concept one has to keep it in its place, and its place is strictly in the preliminary stages of an analytical argument, not in the framing of hypothesis to be tested against the facts, for we know perfectly well that we shall not find facts in a state of equilib-

rium. Yet many writers seem to conceive the long-period as a date somewhere in the future that in equilibrium something is true . . . then it somehow does not matter that every day now and to come, it will not be true. [Robinson, 1962, p. 81]

Pasinetti uses a full-employment equilibrium model "as a logical framework to answer interesting questions about what ought to happen if full employment is to be kept over time, more than as a behavioral theory expressing what actually happens." [Pasinetti, p. 119]

Thus, even though an equilibrium or steady rate of expansion may be described within the post-Keynesian framework, neither is intended to be used as a description of actual activity in the economy. Rather, each of these concepts is intended to be a description of a situation which is meant to be disturbed or changed. Equilibrium or a steady expansion path represents a point of reference with which actual economic activity can be compared.

The importance of the distinction between the uses of equilibrium analysis and the notion of causality for the post-Keynesians, as distinct from the neoclassical view, can only be understood within the context of the Keynesian research program. With the belief that the economy does not automatically reach toward a market-clearing equilibrium, the purpose of economic analysis is to explain change or movement of the system. Furthermore, the forces causing change may be endogenous to the system. If a model of that system is completely determined and causality is embodied in the simultaneity of determination, any explanation of change must be due to variables *not* included in the system, i.e., exogenous shocks. Yet, the Keynesian and the post-Keynesian view are that instability and fluctuations result from the normal operation of a developed economy. Thus, an interesting question to Keynes and to the post-Keynesians is the explanation of this endogenously created change.

THE APPROACHES COMPARED

The purpose of this paper is to compare the systematic framework of the post-Keynesians and neoclassical research programs. I have stated that the two programs are rivals in the sense that the "hard core" associated with each program contradicts the other. It is my contention that the Keynesian research program was originally "grafted on" to the neoclassical program. Both originated from essentially the same works. Both accepted the principle that man is

"rational"—that he is teleologically oriented. The primary difference in the "hard cores" is the assumption that man can have complete knowledge with which to behave "rationally." The effect of this assumption is on the degree of coordination possible in a system where action is a result of "rational" individual choices. This placed the emphasis of the analysis directly on the macroeconomic level. The problem to be analyzed involved the degree of coordination of the whole economic process, not the individual actions. This made the Keynesian program appear to be applicable only to the aggregate level of economic activity. Since neoclassical theory up to that time had been primarily concerned with the analysis of individual actions or choices, the inconsistency between the two programs was not immediately obvious.

The reactions to the new Keynesian research program were mixed. The "conservative" reaction, as it is termed by Lakatos, was to begin a reinterpretation of the Keynesian program along neoclassical lines. [Lakatos, p. 144] This began the process that produced the "neoclassical-Keynesian" synthesis.

While Keynes was still attempting to refine this new way to viewing economic phenomena . . . a number of other economists were trying to comprehend Keynes' ideas in terms of traditional theory. Hicks published a "potted version" of what he believed to be Keynes' central argument, using the famous IS-LM diagram of a general equilibrium system. This began a retrograde movement of modification, alteration, and distortion of the new paradigm to force it into the older neoclassical mold. [Davidson, pp. 276-77]

The development of these so-called "Keynesian" macroeconomic models indicates that better explanation involves the inclusion of more and more variables in a simultaneously determined model. Criticisms of the Samuelson-Hanson "cross" include the nondetermination of the price level and the insufficient treatment of the money-market. [Weintraub, pp. 45-46] The IS-LM framework was considered an improvement over the "cross" as it made equilibrium in the money market and equilibrium in the commodity market simultaneously determined. Thus, we see the development of "Keynesian" economics as a movement toward more comprehensive models exhibiting simultaneity of determination. The IS-LM framework, however, was still unable to show determination of the price level. To present a "plausible theory of inflation" the "Keynesians" had to

refer to the Phillips curve. The Patinkin model, which is said to illustrate the Keynesian-neoclassical synthesis, explains the price level by making it also determined in the general equilibrium model. Minsky summarizes the development of these "Keynesian" models as follows:

The journey through various standard models that embody elements derived from *The General Theory* has led us to the position that such Keynesian models are either trivial (the consumption-function models), incomplete (the IS-LM models without a labor market), inconsistent (the IS-LM models with a labor market but no real-balance effect), or indistinguishable in their results from those of the older quantity-theory models (the neoclassical synthesis). [Minsky, 1975, p. 53]

The Patinkin model is said to illustrate the neoclassical-Keynesian synthesis because non-clearing in the labor market is assumed possible only if money wages are rigid; or, in the Patinkin analysis, if we assume such behavior in the labor market, the labor market is said to be in disequilibrium, and adjustment to full-employment will be delayed. [Patinkin, Ch. 13] The notion is that wage-rigidity is "responsible for unemployment, involuntary unemployment disappears in the neoclassical world." [Minsky, 1975, p. 53] This conflicts with Keynes' view that money wage stability gives price stability and that wage flexibility might make things worse.

My purpose here is not to give a complete summary of current neoclassical macrotheory, but to show that the method involves assuming a full-employment equilibrium and analysis of the economy in terms of movement toward that ideal position. What are given as "Keynesian" contributions to involuntary unemployment are those things which either prevent the automatic adjustment back to equilibrium or prolong the adjustment period. Patinkin, rather than distorting the labor supply curve of his model with the assumption that labor has "money illusion" (rigid wages), simply refers to such behavior as sub-optimal. Thus, involuntary unemployment is a disequilibrium situation. The implication is that if labor would behave "optimally" or "rationally," full employment would automatically be achieved.

This is the real impact of the innocuous tautology of the preceding section, not that involuntary unemployment can be defined away, but that it can have no meaning within the confines of static equilibrium analysis. Conversely, the es-

sence of dynamic analysis is involuntariness: its domain consists only of positions *off* the demand or supply curves. Indeed, it is this very departure from these curves, and the resulting striving of individuals to return to the optimum behavior which they represent, which provides the motive power of the dynamic process itself. [Patinkin, p. 323]

This is the primary distinction between the approach of the neoclassical research program and the post-Keynesian research program. The belief that "rational" behavior can and does lead to a coordinated or efficient outcome which is the "ideal" position for society is associated with the belief that the movement of the whole economic process can be analyzed as the return to that optimum. If the dynamic process can be explained as a movement toward equilibrium or the optimum, then forces that create instability or disequilibrium must be exogenous to the system. Instability must come from outside the normal operation of the economic process, and instability is explained as a result of interference in the otherwise smooth-functioning free market system.

As set out above, the primary characteristics of the post-Keynesian approach are, first, to look for the explanation of instability as endogenous to the operation of an economic process, and second, to explain the movement of an economic process in terms of variables, endogenous and exogenous, which produce change. They do not associate mathematical determinism of a model with causality or situational determinism in the economic process. Most importantly, they do not assume the system automatically moves toward the "efficient" or "ideal" level of output in the neoclassical sense. Their purpose is to "explain" the actual level of economic activity, not to ask what is preventing that activity from reaching some Pareto optimal position.

THE METHODOLOGICAL ISSUES

Accepting that two contradictory research programs do exist, what is the nature of the debate between these two groups? The debate goes on in terms of demarcation rules for accepting specific theories. As such rules are methodologically determined within the context of a given research program, one would not expect to find agreement between the two groups. We must look for what is accepted by each group as predictive power and explanation. Each group should be able to argue that their particular program is more "progressive" or better "explanation" than the other based on the methodological

decision within each program as to what constitutes such "progressivity."

One of the major criticisms of neoclassical theory by the post-Keynesians is that the theory lacks contact with the "real world." Because the "hard core" of the post-Keynesian research program involves the belief that prereconciled coordination is not possible, a theory presented in these terms is unacceptable to them as an "explanation" of the real economic process. Under the influence of rules of demarcation described as naive methodological falsificationism, they reject equilibrium theory as a "metaphysical" theory: i.e., an equilibrium automatically obtained is a statement which is in direct conflict with an "observation" they treat as "unproblematical background knowledge"—man is *not* capable of perfect knowledge. Thus, equilibrium models and conclusions drawn from equilibrium models cannot and do not apply to the "observable" world.

One neoclassical defense against this criticism is that "explanation" is "prediction." The argument is that the assumptions of the model or theory need not be "realistic." The test of a good theory is how well it predicts.

Complete "realism" is clearly unattainable, and the question whether a theory is realistic "enough" can be settled only by seeing whether it yields predictions that are good enough for the purpose at hand or that are better than predictions from alternative theories. Yet the belief that a theory can be tested by the realism of its assumptions independently of the accuracy of its predictions is widespread and the source of much of the perennial criticism of economic theory as unrealistic. Such criticism is largely irrelevant . . . [Friedman, p. 41]

Prediction is not defined, but it is apparently taken as probabilistic predictions of particular observed relationships with rules agreed upon within the neoclassical research program concerning what is acceptable as "good enough." It should be noted, however, that the model from which observable conclusions are deduced is an equilibrium model with the simultaneity of determination related to the notion of causality. So, it could be argued that "explanation" includes the need to develop what is in the neoclassical view as a single consistent treatment of the phenomena to be "explained."

It is also argued by some that a particular set of assumptions should be chosen over alternatives "on the grounds of the resulting economy,

clarity, and precision in presenting the hypothesis." [Friedman, p. 40]
Lakatos refers to this as the methodology of simplicism.

I use this rather ugly term for methodologies according to which one cannot decide between theories on empirical grounds: a theory is better than another if it is simpler, more "coherent," more "economical" than its rival. [Lakatos and Zahar, p. 361]

It is important, of course, that the hypothesis suggest some implication that can be "readily checked with observation" or that they bring out some "connection with other hypotheses dealing with related phenomena." [Friedman, p. 41]

The important point is this: Consistent with the view that an equilibrium or optimal state automatically results from the "rational" actions of individuals, the driving force of the economic process is assumed to be the movement toward that optimum. Thus, in the neoclassical research program, an adequate or acceptable "explanation" of the actual movement of the economic process involves the explanation of this movement toward equilibrium. The "predictions" by which the theory is appraised consist of conclusions derived from the equilibrium model. Assuming a change in one of the variables, how must the other variables change in order to restore equilibrium? Instability is not due to individuals behaving "rationally" and can only be "explained" as the influence of exogenous shocks to the system. It is accepted that the economy is not at equilibrium and may never reach equilibrium due to repeating exogenous shocks. How good "predictions" must be in order to be "good enough" is a methodological decision within the research program. And, there is by no means agreement even within the neoclassical camp with regard to these methodological issues.

The post-Keynesian research program involves the "hard core" belief that an economic process organized by "rational" individual decision-making does not in principle automatically move toward coordinated optimum as described by full-employment equilibrium. While accepting the notion that individuals attempt to maximize their position, they reject the notion that such actions automatically result in a well-coordinated or optimal economic state. Indeed, "rational" individual decision-making may create instability. Therefore, the driving force or actual movement of the economic process cannot be "explained" as movement toward that equilibrium, and conclusions

regarding relationships between variables and causality as they exist in the economic world cannot be derived from the solution of the equilibrium model. Their "empirical" observations and "predictions" are derived from models which do not require completeness in the form of market-clearing equilibrium. How good their "predictions" are should also be appraised within the context of their own research program. And, the "explanation" of actual movement of the economic process involves the analysis of "rational" changes in behavior when expectations are not realized (endogenous source) and technological and physical constraints on long-run development (exogenous source). A primary endogenous source of instability is discussed as the fluctuation on the expected return from capital which results from changes in the level of the capital stock.

CONCLUSION

In conclusion, we must agree that at this time no *objective* way to choose between these two alternative programs may exist. Such a choice at the very least, requires some consensus as to what constitutes a better theory in terms of predictive and explanatory factors. The debate between the neoclassical and the post-Keynesian economists has been primarily in terms of rules of demarcation between theories. However, no agreement regarding what those rules should be has been reached even among members of the same program. It is not clear what the members of each of these programs would accept as a valid test of their theories. Debates on theoretical issues between members of the competing programs are meaningless as the terms are defined within the context of the two different research programs.

While there is substantial agreement about what constitutes the empirical base, the data being gathered from national income accounting, there are irreconcilable differences in the accepted framework or view of the world associated with each research program.⁵ With no consensus concerning what constitutes explanation or pre-

⁵ It should be emphasized that these irreconcilable differences apply primarily to macroeconomic theory. Neoclassical models developed in areas such as industrial organization in which the assumption of maximizing behavior is used without the assumptions of perfect competition, can be said to be more consistent with the post-Keynesian program than with the neoclassical program. The post-Keynesians accept the notion that man is "rational" in the sense that he attempts to maximize his net benefits. They reject the notion that such behavior on the part of all decision makers results in an automatically coordinated optimal state which is "best" for society.

diction, there is no consensus concerning which view is the appropriate approach for understanding the economic process. Those who believe the economy to be inherently stable and well coordinated, with instability a result of exogenous shocks preventing the system from reaching the implied "ideal," tend to fall in the neoclassical camp. Those who believe that a free-market economic process is not inherently stable, but generates within itself forces which create instability and fluctuations in the level of economic activity, will tend to side with the post-Keynesians.

Perhaps the only resolution to the debate is to adopt *subjective* criteria or criteria outside the theoretical framework based on the ability of the different approaches to meet the needs and goals of society. "The respect accorded to a discipline depends in the last resort on its power to serve society." [Mishan, p. xvii] The normative influences in neoclassical welfare theory have been made explicit. Even though both neoclassical and post-Keynesian macrotheory may attempt to be "value free," when policy prescriptions are made, the implicit beliefs as to what is "best" for society are involved. Exposing the "hard core" elements of the competing research programs may make the underlying beliefs explicit and, therefore, pave the way for discussion from which new criteria may emerge.

REFERENCES

- Arrow, Kenneth J. and F. H. Hahn. *General Competitive Analysis*, San Francisco, 1979.
- Asmakopulos, A. "Post-Keynesian Growth Theory," in *Modern Economic Thought*, edited by Sidney Weintraub, College Park PA, 1977.
- Blaug, Mark. *The Methodology of Economics*, Cambridge, 1980.
- Davidson, Paul. "Post-Keynesian Monetary Theory and Inflation," in *Modern Economic Thought*, edited by Sidney Weintraub, College Park PA, 1977.
- Eichner, Alfred S. and J. A. Kregel. "An Essay on Post-Keynesian Theory: A New Paradigm in Economics," *Journal of Economic Literature*, 13, December 1975, pp. 1293-1314.
- Keynes, John M. *Economic Articles and Correspondence (Towards the General Theory)*, Vol. XIII of *The Collected Writings of John Maynard Keynes*, edited by Donald Moggridge, London, 1973a.
- . *The General Theory of Employment, Interest, and Money*, New York, 1965.
- . *Social, Political, and Literary Writings*, Vol. XIV of *The Collected Writings of John Maynard Keynes*, edited by Donald Moggridge, London, 1973b.
- . *Treatise on Probability*, Vol. VIII of *The Collected Writings of John Maynard Keynes*, edited by Donald Moggridge, London, 1973c.
- Kregel, J. A. "Some Post-Keynesian Distribution Theory," in *Modern Economic*

- Thought*, edited by Sidney Weintraub, College Park PA, 1977.
- Lakatos, Imre. "Falsification and the Methodology of Scientific Research Programmes," in *Criticism and the Growth of Knowledge*, edited by Imre Lakatos and Alan Musgrave, Cambridge, 1970.
- and Elie Zahar. "Why Did Copernicus' Research Program Supersede Ptolemy's?" in *The Copernican Achievement*, edited by Robert S. Westman, Berkeley, 1975.
- Latsis, Spiro J. "A Research Programme in Economics," in *Method and Appraisal in Economics*, edited by Spiro Latsis, Cambridge, 1976.
- Leijonhufvud, Axel. "Schools, 'Revolutions' and Research Programmes in Economic Theory," in *Method and Appraisal in Economics*, edited by Spiro Latsis, Cambridge, 1976.
- Minsky, Hyman. "An Economics of Keynes' Perspective on Money," in *Modern Economic Thought*, edited by Sidney Weintraub, College Park PA, 1977.
- . *John Maynard Keynes*, New York, 1975.
- Mishan, E. J. *Introduction to Normative Economics*, New York, 1981.
- Pasinetti, Luigi L. *Growth and Income Distribution*, Cambridge, 1974.
- Patinkin, Don. *Money, Interest and Prices*, 2nd ed., New York, 1965.
- Robinson, Joan. *Economic Philosophy*, Chicago, 1962.
- . "What Are the Questions?" *Journal of Economic Literature*, December 1977.
- Toulmin, Stephen E. "Commentary," in *The Copernican Achievement*, edited by Robert S. Westman, Berkeley, 1975.
- Weintraub, Sidney. "Hicksian Keynesianism: Dominance and Decline," in *Modern Economic Thought*, edited by Sidney Weintraub, College Park PA, 1977.